Pneumatic Press



Category: **Power Tools & Workshop Machines** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Pneumatic press

What it is used for

A pneumatic press uses compressed air to apply force for pressing, fitting or forming suitable components. Safe operation requires correct pressure settings, secure workholding and protection from moving press parts.

Main parts

- Identify the press frame, bed and pressing head.
- Locate the pneumatic cylinder that converts compressed air into movement.
- Identify the ram or pressing tool that contacts the workpiece.
- Locate the compressed-air inlet, hose and regulator.
- Identify the pressure gauge and air control valve.
- Locate the operating pedal or two-hand control, depending on design.
- Identify guards, emergency stop and air-isolation valve.

Before you start

1. Read the operating instructions and receive supervised training before use.
2. Inspect the frame, ram, air hose, fittings and guards for damage.
3. Check that the compressed-air supply matches the manufacturer's specifications.
4. Confirm the pressure regulator and gauge operate correctly.
5. Select suitable pressing tooling and secure the workpiece on the bed.
6. Keep hands clear of the pressing zone and adjust the guard as required.
7. Remove loose objects and ensure the emergency stop is accessible.

How to use

1. Isolate the air supply before installing tooling or adjusting the press.
2. Fit the correct pressing tool and secure it according to instructions.
3. Position the workpiece centrally and support it firmly on the press bed.
4. Set the working pressure within the approved operating range.
5. Reconnect the air supply and check for leaks or abnormal sounds.
6. Close the guards and select the designated operating control.
7. Activate the press using the approved pedal or control arrangement.
8. Observe the pressing cycle from a safe position without reaching into the work area.
9. Release the control and wait until the ram returns to its safe position.
10. Isolate the air supply before removing the finished workpiece or changing settings.

⚠ Safety

- Never place fingers or hands between the ram and press bed.
- Never exceed the rated pressure or load capacity of the press.
- Keep guards and safety controls functional; never bypass them.
- Isolate compressed air and release trapped pressure before servicing.
- Never use damaged hoses, loose fittings or an unstable press frame.
- Stop operation immediately if the press moves unexpectedly or leaks air.

Care & storage

- Isolate air and release stored pressure before cleaning or maintenance.
- Clean the press bed and remove debris after each session.
- Inspect hoses, fittings, ram and guards regularly for wear or leaks.
- Follow manufacturer instructions for lubrication and scheduled maintenance.

Common mistakes to avoid

- Positioning the workpiece off-centre and causing unstable pressing.
- Increasing air pressure beyond the specified operating range.
- Reaching into the pressing zone before the ram has stopped.
- Adjusting tooling while compressed air remains connected.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

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